

Ravnoor Singh

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EDUCATION

Wright State University

M.SC. IN COMPUTER SCIENCE

Dayton, Ohio, USA

August 2024 – December 2026

Guru Nanak Dev University

B.TECH. IN COMPUTER SCIENCE ENGINEERING

Amritsar, Punjab, India

September 2020 – May 2024

TECHNICAL SKILLS

Languages/Software: C++, Java, Python, C#, JavaScript, R, Bash, ASP.NET, HTML, CSS, Linux, Quarto

Web Frameworks: React, Node.js, Express.js, Material-UI, RapidAPI, EJS, Next.js

Machine Learning & AI: PyTorch, TensorFlow, LangChain, Hugging Face, Retrieval-Augmented Generation (RAG), scikit-learn, LlamaIndex

Big Data & Databases: SQL, MongoDB, Tableau, MySQL, Vector Databases, Hadoop, PySpark, Kafka, ETL Pipelines

Cloud & DevOps: Amazon Web Services (AWS), Docker, CI/CD Pipeline

Version Control: Git, GitHub, DVC, DagsHub

WORK EXPERIENCE

Full Stack Engineer Intern

THE COMMONS XR

Remote, USA

May 2025 – August 2025

- **Integrated backend services** to reflect and manage the status of core infrastructure, including Virtual Machines and Azure-based serverless functions on the frontend.
- Contributed to **frontend bug resolution** by debugging and refining filtering, navigation, and admin functionalities

Section Leader

CODE IN PLACE

Remote, USA

April 2025 – June 2025

- Led weekly live Python sections for a globally distributed cohort of **120+ countries**, guiding learners through Stanford's CS106A curriculum using structured problem-solving and live code walkthroughs.
- Mentored learners **1:1** through debugging, algorithmic thinking, and project work, accelerating comprehension and retention through interactive exercises and real-world examples.

Undergraduate Research Assistant

DEPARTMENT OF COMPUTER ENG. & TECHNOLOGY, GURU NANAK DEV UNIVERSITY

Amritsar, Punjab, India

January 2024 – June 2024

- Delivered a production-grade ML system for early disease prediction, training and validating **10+ models on 187K+** patient records with rigorous preprocessing and imbalance handling.
- Achieved up to **99.3%** accuracy and **99%** recall by designing a novel ensemble voting framework, reducing false negatives in high-risk medical predictions.
- Shipped research to real impact, publishing 2 research papers in Nature ([Scientific Reports](#)) and ([Franklin Open](#)) paper covering **5 diseases** along with integration of **explainable AI** and demonstrating clinically reliable ML performance.

SDE Intern

DESIQNA

Remote, India

September 2023 – December 2023

- Built and shipped responsive interview-experience workflows using React.js and Tailwind CSS, delivering a consistent UX across desktop and mobile.
- Implemented secure user authentication and Google reCAPTCHA, strengthening platform security and preventing automated abuse.

PROJECTS

ScreenSavvy | REACTJS, EXPRESSJS, NODEJS, MONGODB, HTML, CSS, MAILTRAP, CLOUDINARY, POSTMAN

- Built and shipped a secure full-stack media platform with JWT authentication, role-based access, and custom REST APIs, supporting **400+ media assets**.
- Integrated Cloudinary-based media pipelines and admin tooling, improving upload reliability and cutting moderation **effort by 40%**.

Distributed TSP Solver | AWS, LINUX, PYTHON, HPC, CONTAINERS, AUTOMATION

- Designed and delivered a fully automated distributed computing pipeline orchestrating **3 compute environments** (AWS, university HPC, Ohio Supercomputer) via a single scripted control node, coordinating job submission, file transfers, and execution across **150 total runs** (15 batches of 10 trials).
- Achieved up to **4x runtime speedup** through parallel execution and dynamic batching, evaluating millions of TSP paths to produce the best-known solution for a **1,090-city U.S dataset**.